

# Finger-Pointing Interface for Human Gesture Recognition Based on Real-Time Geometric Comprehension

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## Abstract

This study proposes a system to recognize finger-pointing gestures and estimate their 3D coordinates to provide an intuitive and natural interface in the field of human-computer interaction. The study delineates a method commonly used in human conversations to describe locations and adapts it for interface application. It focuses on pose estimation using a stereo camera and provides mathematical and technical explanations, demonstrating the application of this interface to a four-wheeled mobile robot. The interaction between the user and the robot is detailed, alongside demonstrations and elucidations of the errors and their causes during the process. The study proposes methods to minimize these errors and explores additional potential applications of the finger-pointing gesture interface, highlighting its feasibility.

## ACM Reference Format:

Minjae Lee, Jiho Bae, Sang-Min Choi, and Suwon Lee. 2024. Finger-Pointing Interface for Human Gesture Recognition Based on Real-Time Geometric Comprehension. In *SIGGRAPH Asia 2024 Posters (SA Posters '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3681756.3697892>

## 1 Introduction

In recent years, technologies that understand and respond to users' intuitive gestures have gained significant attention in human-computer interaction (HCI). Research is actively conducted to improve interactions between users and computer systems through simple gestures, like finger-pointing. These technologies can enhance user experience by offering more natural and efficient interfaces. Unlike traditional use of coordinates ( $X, Y, Z$ ), humans describe locations intuitively through gestures, often using pronouns like "this" or "that." Such communication is natural but poses challenges for computer systems. Traditional HCI systems rely on physical devices like mice or keyboards, which are restrictive and less intuitive. This study aims to develop an interface that estimates 3D coordinates through finger-pointing gestures using a stereo camera. The gesture interface helps users express their intentions and improves the system's efficiency, making HCI more natural and effective.

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SA Posters '24, December 03–06, 2024, Tokyo, Japan

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ACM ISBN 979-8-4007-1138-1/24/12

<https://doi.org/10.1145/3681756.3697892>

## 2 Methodologies

### 2.1 Hardware Configuration

Since monocular cameras cannot calculate depth information, the introduction of stereo cameras is necessary to build an interface based on 3D coordinates for finger-pointing gestures. Before using stereo cameras, camera calibration must be performed to obtain intrinsic parameters for each camera and extrinsic parameters between the cameras. A fixed-focus stereo camera was used to ensure the accuracy and consistency of finger-pointing gestures.

### 2.2 Geometric Comprehension

$$P_{\tau} = P_S + \|\overrightarrow{P_S P_F}\|^2 \cdot \overrightarrow{P_S P_F} \cdot \overrightarrow{P_S P_W}^{-1} \cdot \overrightarrow{P_S P_W} \quad (1)$$

Images  $I_L$  and  $I_R$  collected from the left and right cameras of the stereo camera system are utilized, with  $I_L$  being used to estimate the human pose. This study introduces MediaPipe[Lugaresi et al. 2019] to estimate 2D coordinates of 33 pose landmarks. To calculate 3D coordinates, corresponding 2D coordinate pairs from  $I_L$  and  $I_R$  are required. As noted, only  $I_L$  is used for pose estimation because MediaPipe Pose does not provide an adequate matching pair with high accuracy. To ensure semantic consistency between the 2D coordinates of pose landmarks from  $I_L$  and  $I_R$ , a disparity map ( $D_L$ ) for the left camera is calculated. The depth values  $Z_C$  for the 33 pose landmarks are extracted from  $D_L$ , and the corresponding  $X_C$  and  $Y_C$  are calculated. To geometrically calculate the pointed destination coordinates ( $P_{\tau}$ ), the 3D coordinates of the shoulder ( $P_S$ ), foot ( $P_F$ ), and wrist ( $P_W$ ) are used, following the method described in Equation 1. The left part of Figure 1 depicts this method and shows an example destination calculated using MediaPipe and  $D_L$ , marked with a blue dot.

## 3 Development

### 3.1 Robot Configuration

One example of applying the proposed finger-pointing interface is a four-wheeled mobile robot (FWMR) equipped with a stereo camera. For experimental convenience, a simple FWMR was adopted, as depicted in the right part of Figure 1, rather than a complex humanoid robot. We utilized a Raspberry Pi-based car kit with a stereo camera capable of rotating up to 180° in the yaw direction. When a person points to a destination with their finger, the robot moves to the calculated location using pre-calibrated wheel rotations.

### 3.2 Central Alignment

To achieve the finger-pointing interface, continuous capture of the finger-pointing gesture is essential. However,  $I_L$  may fail to

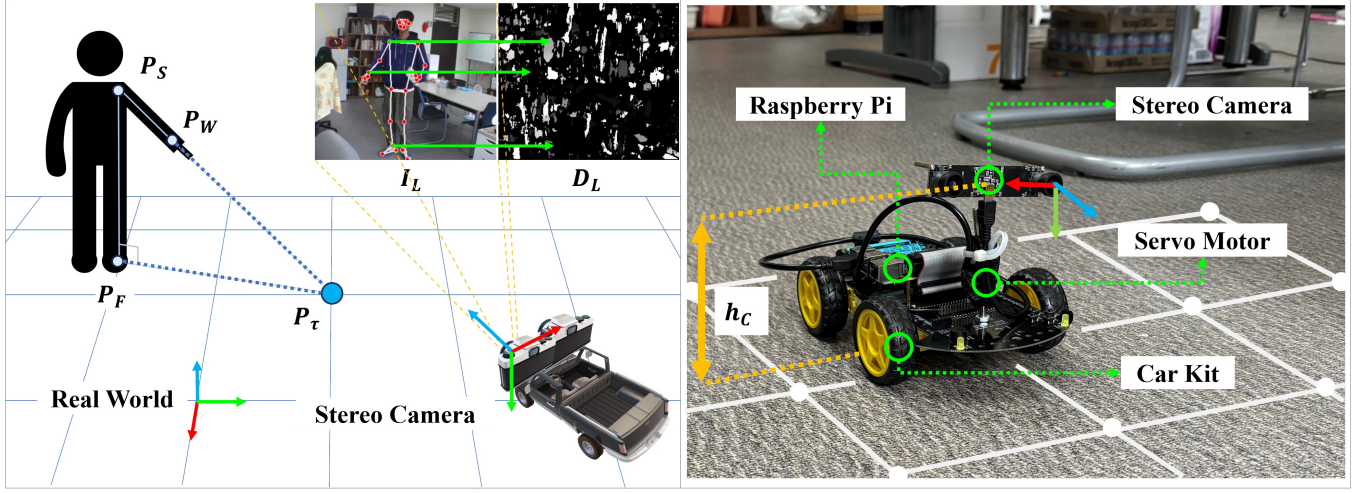


Figure 1: Left: Diagram of geometric comprehension to achieve the target coordinate ( $P_T$ ) using real-time image ( $I_L$ ) and disparity map ( $D_L$ ); Right: Hardware configuration: Raspberry Pi car kit with rotatable stereo camera

capture the gesture due to robot movement or rotation, or human movement. The average 2D coordinates of human pose landmarks  $\psi$  can be calculated as  $\lfloor w \sum_{i=1}^{33} y_i / 99 \rfloor$ . Here,  $w$  represents the width of  $I_L$ , and  $y_i$  is the  $y$ -coordinate of the  $i$ -th landmark. If  $\psi$  equals 0, the stereo camera's motor rotates left; if 1, it remains unchanged; if 2, it rotates right. This average calculation ensures the motor rotates appropriately to continuously capture human gestures.

### 3.3 Continuous Localization

During the initialization and placement of the car robot, the starting point is assumed to be the origin (0, 0, 0) with no rotation. Similarly, the motor attached to the camera is also assumed to be at the origin. When a finger-pointing gesture occurs, the robot moves and rotates to the position calculated in Section 2.2, and the motor attached to the camera rotates simultaneously. The car robot continuously saves its position, rotation angle, and the stereo camera's rotation angle, enabling continuous localization.

## 4 Demonstration, Test, and Discussion

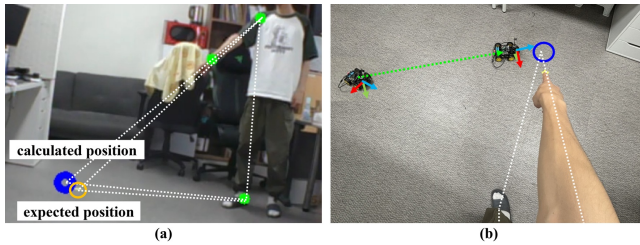


Figure 2: Errors in the robot's view (a) and human's (b).

The car robot moves to the pointed location through the proposed interface. As shown in Figure 2(a) and (b), it is expected to move from the start to the target, but due to errors (human, MediaPipe, hardware, and software), it reaches a different position.

Human error from visual inaccuracies is excluded in this study. To focus on MediaPipe error, a simple experiment was conducted. A grid system was set, and the robot was placed at (0, 0,  $h_C$ ), where  $h_C$  is the camera height. The user pointed at a grid spot, its 3D coordinate was calculated using  $D_L$  and Equation 1, and the robot moved to that point. Figure 2(a) shows the actual target (yellow) and the error point (blue). Figure 2(b) compares the user's target (yellow), the robot's calculated point (blue), and its final position (pink with green arrow), with deviations due to ground material and PWM changes from low battery voltage. These errors accumulate, affecting the robot's position stored wrong. Real-time localization techniques like SLAM can help resolve such issues.

## 5 Conclusion

This study introduced a finger-pointing interface using stereo cameras and MediaPipe Pose, applied to FWMR for target movement. Although successful, discrepancies between expected and actual positions were analyzed. The potential of finger-pointing in HCI and HRI is evident. Future work will focus on enhancing accuracy and reliability, enabling tasks like fetching items, which could reduce loneliness. This system shows promise for single-person households and disabled individuals, improving both physical and emotional interactions with robots.

## Acknowledgments

This work was supported by the "Leaders in Industry-university Cooperation 3.0" Project funded by the Ministry of Education and National Research Foundation of Korea and Korea government (No. RS-2022-00165785).

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